ANURAG SINGH

Applied AI & Analytics · LLMs, RAG & AI Agents · Fintech & Banking

India | anurag1481@gmail.com | linkedin.com/in/anurag-singh-037a83103 | github.com/anurag-devx

SUMMARY

Analytics professional with 6 years building machine learning and GenAI systems in banking and fintech: real-time credit models and NLP support automation at Simpl, and RAG-based AI governance and data-quality tooling at Citi. Creator of offline-agent, an open-source framework for running permission-controlled AI agents on local LLMs. Hands-on across modeling, retrieval and agent design.

TECHNICAL SKILLS

GenAI & LLMs: RAG, AI agents, tool calling, prompt engineering, embeddings, semantic search, hybrid retrieval (FAISS + BM25), cross-encoder reranking, Ollama, LLaMA-3, Qwen, GPT-4o-mini, Sentence Transformers

Machine Learning: XGBoost, NLP intent classification, K-Means, statistical modeling, A/B testing, model evaluation

Engineering & Data: Python, SQL, PySpark, FastAPI, Streamlit, pytest, GitHub Actions (CI), Git, HDFS, Impala, SAS, Tableau

EXPERIENCE

Information Management Officer, Digital Product Strategy | Citi India Dec 2024 – Present

Enterprise AI Adoption team: credit analytics, compliance and data quality across digital banking

- Architected a RAG-based AI governance platform (enterprise LLM, embedding-based retrieval, PySpark/HDFS) with automated ML model review, human-in-the-loop approvals and audit logging for compliance traceability.
- Integrated the platform with GitHub to auto-review model code and commits, flagging governance issues before merge.
- Built a Data Quality & Compliance Framework (PySpark, Impala) that continuously monitors 100+ metrics across 10+ risk reports and flags anomalies before they reach stakeholders.
- Led the SAS-to-Enterprise Analytics Platform migration on HDFS, automating 20+ manual risk-reporting workflows and saving 40+ hours per month.
- Built launch analytics for Paze Wallet and early-adoption analysis for FlexPay (Apple Pay BNPL); received the Citi Silver Gratitude Award for end-to-end Paze Wallet analytics.

Senior Data Analyst, Product & Operations | Simpl (GetSimpl) Dec 2021 – Nov 2024

- Built a real-time gradient-boosted credit limit recommendation system, increasing credit approval rates by 6%.
- Designed a two-phase conversational AI framework, improving intent classification by 15% and cutting support load by 20%.
- Developed predictive risk scoring for behavior-based collections, recovering $1.1M in delinquent accounts within 45 days.
- Built a merchant risk-scoring model (approval rate, refund rate, chargebacks, support tickets) to flag high-risk sub-merchants, leading to revised merchant terms.
- Built an Instant Refund eligibility model and A/B-tested it, cutting delinquency ~20% net of chargeback costs; led pre-launch pricing analytics for Simpl Select.
- Built end-to-end support analytics and took ownership of it, recognized with the 2023 Annual Award (“Do More With Less”).

Operations Analyst, Strategic Operations | ESSI Dec 2020 – Dec 2021

- Built pricing optimization models that improved bid win rate by 30% across 20+ network security projects; applied heuristic optimization to resource allocation, cutting operating expenses by 20%.

PROJECTS

Forward Court | Meta Global AI Developer Hackathon 2026

- Building a multi-agent fact-checker for viral WhatsApp forwards: claim intake, grounded evidence retrieval, prosecutor and defender agents, and a judge agent producing a shareable verdict card.

offline-agent | Open-source framework for AI agents on local LLMs github.com/anurag-devx/offline-agent

- Built a Python framework (~5K lines, MIT, pip-installable, v0.2.0) to design, preview, validate, run and schedule agents on local models via Ollama; agents defined in YAML, with a Streamlit studio and CLI.
- Designed a rule mode where the model makes one decision and code executes and verifies it: on a file-sorting benchmark (qwen3.5:9b, 16 GB laptop) it used 30x fewer tokens (532 vs 15,705) and finished in 89 s vs ~17 min, completing a task the classic tool-calling loop failed.
- Enforced safety in code, not prompts: permission policy, human approvals, a read-only verifier agent, undo and run logs; 41 tests in GitHub Actions CI on 3 OSes.

MarketX | NSE equity research platform github.com/anurag-devx/marketx

- Built a local research app for Indian equities: Angel One, NSE filings, XBRL results, news and macro data feed a multi-agent pipeline (regime, screener, technical, risk, validation) with RAG search over news and filings on local LLMs (Ollama).
- Market Lab predicts 1%+ next-day moves; in walk-forward, point-in-time tests (2019–2026), top-20 picks moved 1%+ 74% of the time vs a 68% base rate, in 8 of 8 years.

Other GenAI projects | Private repositories, available on request

- **Doc2Insight:** document Q&A platform with multi-format ingestion and source-aware RAG answers (GPT-4o-mini).
- **Local-First RAG Assistant:** PDF-grounded Q&A with hybrid retrieval (FAISS + BM25), FastAPI backend and Streamlit UI.

EDUCATION & CERTIFICATION

Generative AI Engineering Professional Certificate | IBM 2026

Supervised Machine Learning | DeepLearning.AI & Stanford Online 2022

MBA, Operations & Statistics | IIIT Allahabad · Silver Medalist 2020

B.Tech, Mechanical Engineering | SPPU Pune · Distinction 2017